

```

#include <iostream>

#include <string>

using namespace std;

/* =====

PERSON 1

Variables + Arrays + Show Rooms

===== */

const int SIZE = 5;

int roomNumber[SIZE] = {101, 102, 103, 104, 105};

bool booked[SIZE] = {false, false, false, false, false};

string customerName[SIZE] = {"", "", "", "", ""};

// Function to show room status
void showRooms()
{
    cout << "\n----- Room Status ----- \n";

    for(int i = 0; i < SIZE; i++)
    {

```

```

        cout << "Room " << roomNumber[i] << " : ";

        if(booked[i] == true)
        {
            cout << "Booked";
        }
        else
        {
            cout << "Available";
        }

        cout << endl;
    }
}

```

```

/* =====

```

PERSON 2

Reserve Room Function

```

===== */

```

```

void reserveRoom()

```

```

{
    int room;
    string name;

```

```
cout << "\nEnter Room Number: ";
```

```
cin >> room;
```

```
for(int i = 0; i < SIZE; i++)
```

```
{
```

```
    if(room == roomNumber[i])
```

```
    {
```

```
        if(booked[i] == false)
```

```
        {
```

```
            cin.ignore();
```

```
            cout << "Enter Customer Name: ";
```

```
            getline(cin, name);
```

```
            customerName[i] = name;
```

```
            booked[i] = true;
```

```
            cout << "Room Reserved Successfully.\n";
```

```
        }
```

```
    else
```

```
    {
```

```
        cout << "Room Already Booked.\n";
```

```
    }
```

```

        return;
    }
}

cout << "Room Not Found.\n";
}

/* =====
PERSON 3
Cancel Reservation Function
===== */

void cancelReservation()
{
    int room;

    cout << "\nEnter Room Number to Cancel: ";
    cin >> room;

    for(int i = 0; i < SIZE; i++)
    {
        if(room == roomNumber[i])
        {
            if(booked[i] == true)

```

```

    {
        booked[i] = false;

        customerName[i] = "";

        cout << "Reservation Cancelled.\n";
    }
    else
    {
        cout << "Room is already empty.\n";
    }

    return;
}

cout << "Room Not Found.\n";
}

/* =====

PERSON 4

Show Reservations + Search Customer

===== */

```

```
// Function to show reservations
void showReservations()
{
    cout << "\n----- Reservations ----- \n";

    for(int i = 0; i < SIZE; i++)
    {
        if(booked[i] == true)
        {
            cout << "Room " << roomNumber[i]
                << " reserved by "
                << customerName[i] << endl;
        }
    }
}
```

```
// Function to search customer
void searchCustomer()
{
    string name;

    cin.ignore();

    cout << "\nEnter Customer Name: ";
```

```
getline(cin, name);
```

```
for(int i = 0; i < SIZE; i++)
```

```
{
```

```
    if(customerName[i] == name)
```

```
    {
```

```
        cout << "Customer Found in Room "
```

```
        << roomNumber[i] << endl;
```

```
        return;
```

```
    }
```

```
}
```

```
cout << "Customer Not Found.\n";
```

```
}
```

```
/* =====
```

```
PERSON 5
```

```
Main Function + Menu + Switch
```

```
===== */
```

```
int main()
```

```
{
```

```
    int choice;
```

```
do
{
    cout << "\n===== Hotel Reservation System =====\n";

    cout << "1. Show Rooms\n";
    cout << "2. Reserve Room\n";
    cout << "3. Cancel Reservation\n";
    cout << "4. Show Reservations\n";
    cout << "5. Search Customer\n";
    cout << "6. Exit\n";

    cout << "Enter Your Choice: ";

    cin >> choice;

    switch(choice)
    {
        case 1:
            showRooms();
            break;

        case 2:
            reserveRoom();
            break;
```

case 3:

cancelReservation();

break;

case 4:

showReservations();

break;

case 5:

searchCustomer();

break;

case 6:

cout << "Program Ended.\n";

break;

default:

cout << "Invalid Choice.\n";

}

} while(choice != 6);

return 0;

}